Orthogonality and change of basis

$$V = \text{Span}\left(\begin{bmatrix}\vec{v}_1\end{bmatrix}, \begin{bmatrix}\vec{v}_2\end{bmatrix}\right)$$

1. Direction
2. ~~Length~~

Pick better vectors

$$V = \text{Span}\left(\begin{bmatrix}\vec{u}_1\end{bmatrix}, \begin{bmatrix}\vec{u}_2\end{bmatrix}\right) \longleftarrow \text{Change of basis}$$

Orthogonal complements

$$\vec{v}\cdot\vec{x}=0$$

$\mathbb{R}^n$ $\quad V=\{\vec{v}_1,\vec{v}_2,\vec{v}_3,\dots\vec{v}_n\}$

$\mathbb{R}^n$ $\quad V^{\perp}=\{\vec{x}_1,\vec{x}_2,\vec{x}_3,\dots\vec{x}_m\}$

$$V^{\perp}=\{\vec{x}\in\mathbb{R}^n \mid \vec{x}\cdot\vec{v}=0 \text{ for every } \vec{v}\in V\}$$

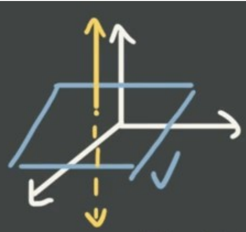
$$\vec{x}_1\cdot\vec{v}=0 \qquad \vec{x}_1\cdot\vec{v}+\vec{x}_2\cdot\vec{v}=0$$

$$\vec{x}_2\cdot\vec{v}=0 \qquad (\vec{x}_1+\vec{x}_2)\cdot\vec{v}=0$$

$$c\vec{x}_1\cdot\vec{v}=0 \qquad c(\vec{x}_1\cdot\vec{v})=0 \qquad c(0)=0$$

$$0=0$$

$$V=\text{Span}\left(\begin{bmatrix}1\\0\\2\end{bmatrix},\begin{bmatrix}-1\\1\\3\end{bmatrix}\right)$$



$$V^{\perp}=\left\{\vec{x}\in\mathbb{R}^3 \mid \vec{x}\cdot\begin{bmatrix}1\\0\\2\end{bmatrix}=0 \text{ and } \vec{x}\cdot\begin{bmatrix}-1\\1\\3\end{bmatrix}=0\right\}$$

$$\vec{x}=(x_1,x_2,x_3)$$

$$V^{\perp}=\text{Span}\left(\begin{bmatrix}-2\\-5\\1\end{bmatrix}\right)$$

$$1x_1+0x_2+2x_3=0$$

$$-1x_1+1x_2+3x_3=0$$

$$\left[\begin{array}{ccc|c}1&0&2&0\\-1&1&3&0\end{array}\right]$$

$$x_1=-2x_3$$

$$x_2=-5x_3$$

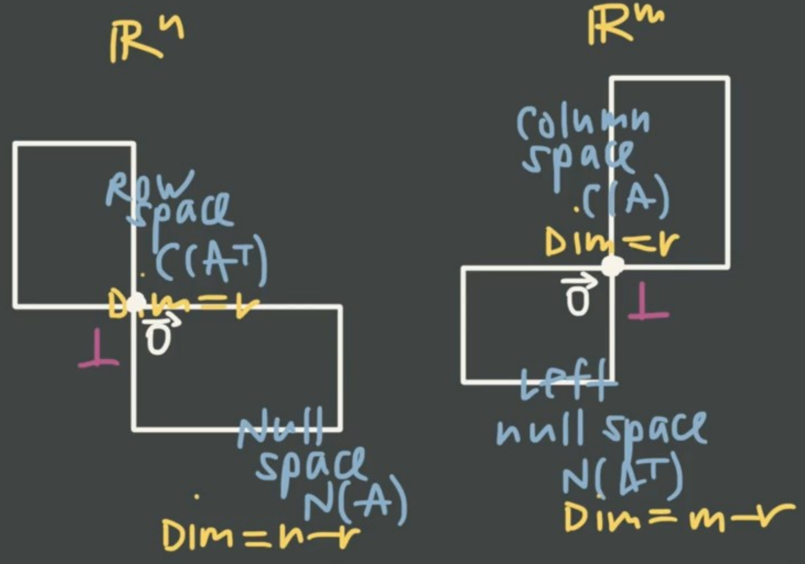
$$\begin{array}{cccc} x_1 & x_2 & x_3 & \\ \end{array}$$
$$\left[\begin{array}{ccc|c}\boxed{\begin{matrix}1\\0\end{matrix}}&\boxed{\begin{matrix}0\\1\end{matrix}}&\boxed{\begin{matrix}2\\5\end{matrix}}&\begin{matrix}0\\0\end{matrix}\end{array}\right]$$

$$\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}=x_3\begin{bmatrix}-2\\-5\\1\end{bmatrix}$$

$$x_1+2x_3=0$$

$$x_2+5x_3=0$$

Orthogonal complements of the fundamental subspaces



$$N(A) = (C(A^T))^{\perp}$$
$$(N(A))^{\perp} = C(A^T)$$

$$C(A) = (N(A^T))^{\perp}$$
$$N(A^T) = (C(A))^{\perp}$$

$\mathbb{R}^n$: V $V^{\perp}$

$$\text{Dim}(V) + \text{Dim}(V^{\perp}) = n$$

$n = 4$, $3 = m$

$$A = \begin{bmatrix} 1 & 0 & 0 & 2 \\ -1 & 1 & 2 & 0 \\ 2 & 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & -1 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 3 \end{bmatrix} = \begin{bmatrix} \boxed{\begin{matrix}1\\0\\0\end{matrix}} & \boxed{\begin{matrix}0\\1\\0\end{matrix}} & \boxed{\begin{matrix}0\\0\\1\end{matrix}} & \boxed{\begin{matrix}2\\-4\\3\end{matrix}} \end{bmatrix}$$

$C(A) = \text{span}\left(\begin{bmatrix} \vdots \end{bmatrix} \begin{bmatrix} \vdots \end{bmatrix} \begin{bmatrix} \vdots \end{bmatrix}\right)$

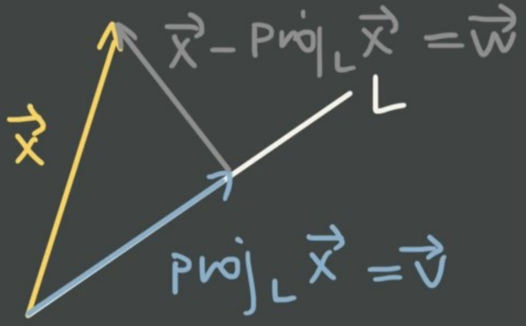
$N(A) = \text{Span}\left(\begin{bmatrix} \vdots \end{bmatrix}\right)$

$C(A^T) = \text{span}\left(\begin{bmatrix} \vdots \end{bmatrix} \begin{bmatrix} \vdots \end{bmatrix} \begin{bmatrix} \vdots \end{bmatrix}\right)$

$N(A^T) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

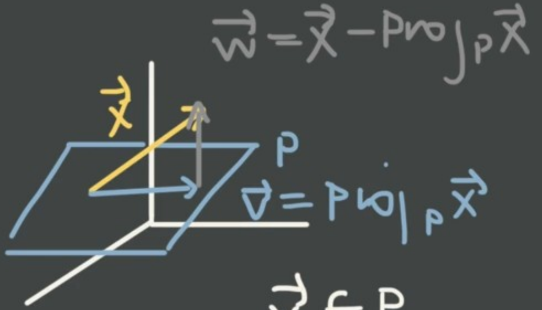
space		Dim
$C(A)$	$\mathbb{R}^3$	3
$N(A)$	$\mathbb{R}^4$	1
$C(A^T)$	$\mathbb{R}^4$	3
$N(A^T)$	$\mathbb{R}^3$	0

Projection onto the subspace



$\vec{v} \in L$

$\vec{w} \in L^{\perp}$



$\vec{v} \in P$

$\vec{w} \in P^{\perp}$

V in $\mathbb{R}^n$

$$\text{proj}_V \vec{x} = \underline{A(A^TA)^{-1}A^T}\vec{x}$$

$$V = \text{span}\left(\begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}\right)$$

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 0 \\ 1 & -1 \end{bmatrix} \qquad A^T = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 0 & -1 \end{bmatrix}$$

$$\left[\begin{array}{cc|cc} 1 & 0 & \frac{2}{11} & -\frac{1}{11} \\ 0 & 1 & -\frac{1}{11} & \frac{6}{11} \end{array}\right]$$

$$\begin{bmatrix} 2 & 1 \\ 1 & 0 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} \frac{2}{11} & -\frac{1}{11} \\ -\frac{1}{11} & \frac{6}{11} \end{bmatrix}\begin{bmatrix} 2 & 1 & 1 \\ 1 & 0 & -1 \end{bmatrix}$$

$$\begin{bmatrix} \frac{4}{11}-\frac{1}{11} & -\frac{2}{11}+\frac{6}{11} \\ \frac{2}{11}+0 & -\frac{1}{11}+0 \\ \frac{2}{11}+\frac{1}{11} & -\frac{1}{11}-\frac{6}{11} \end{bmatrix}\begin{bmatrix} 2 & 1 & 1 \\ 1 & 0 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 3/11 & 4/11 \\ 2/11 & -1/11 \\ 3/11 & -7/11 \end{bmatrix}\begin{bmatrix} 2 & 1 & 1 \\ 1 & 0 & -1 \end{bmatrix}$$

$$\begin{bmatrix} \frac{6}{11}+\frac{4}{11} & \frac{3}{11}+0 & \frac{3}{11}-\frac{4}{11} \\ \frac{4}{11}-\frac{1}{11} & \frac{2}{11}-0 & \frac{2}{11}+\frac{1}{11} \\ \frac{6}{11}-\frac{7}{11} & \frac{3}{11}-0 & \frac{3}{11}+\frac{7}{11} \end{bmatrix} = \begin{bmatrix} \frac{10}{11} & \frac{3}{11} & -\frac{1}{11} \\ \frac{3}{11} & \frac{2}{11} & \frac{3}{11} \\ -\frac{1}{11} & \frac{3}{11} & \frac{10}{11} \end{bmatrix}\vec{x} = \frac{1}{11}\begin{bmatrix} 10 & 3 & -1 \\ 3 & 2 & 3 \\ -1 & 3 & 10 \end{bmatrix}\vec{x}$$

Least squares solution

$\vec{x}^* = \text{Proj}_{C(A)}\vec{b}$

$A\vec{x} = \vec{b}$

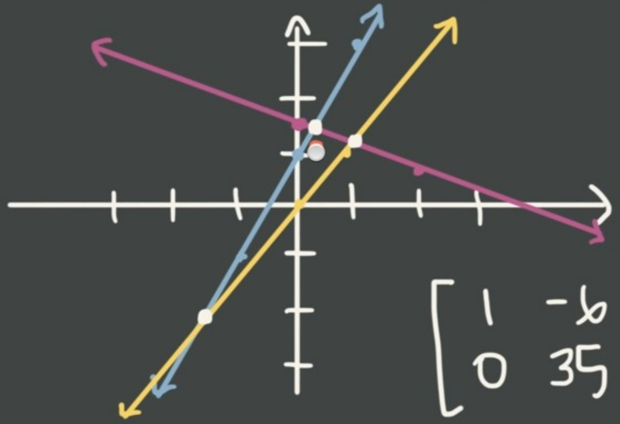
$$\begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

$$x_1\begin{bmatrix} 2 \\ 1 \end{bmatrix} + x_2\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

$2 \qquad 1 \qquad \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

$A^TA\vec{x}^* = A^T\vec{b}$

$-2x+y=1$	$y=2x+1$
$-x+y=0$	$y=x$
$x+2y=3$	$2y=-x+3$
	$y=-\frac{1}{2}x+\frac{3}{2}$



$$\begin{array}{ccc} A & \vec{x} & \vec{b} \\ \begin{bmatrix} -2 & 1 \\ -1 & 1 \\ 1 & 2 \end{bmatrix} & \begin{bmatrix} x \\ y \end{bmatrix} = & \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} \end{array}$$

$$\begin{bmatrix} 6 & -1 \\ -1 & 6 \end{bmatrix}\vec{x}^* = \begin{bmatrix} 1 \\ 7 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} 6 & -1 & 1 \\ -1 & 6 & 7 \end{array}\right] \to \left[\begin{array}{cc|c} 1 & -6 & -7 \\ 6 & -1 & 1 \end{array}\right]$$

$$\left[\begin{array}{cc|c} 1 & -6 & -7 \\ 0 & 35 & 43 \end{array}\right] \to \left[\begin{array}{cc|c} 1 & -6 & -7 \\ 0 & 1 & \frac{43}{35} \end{array}\right]$$

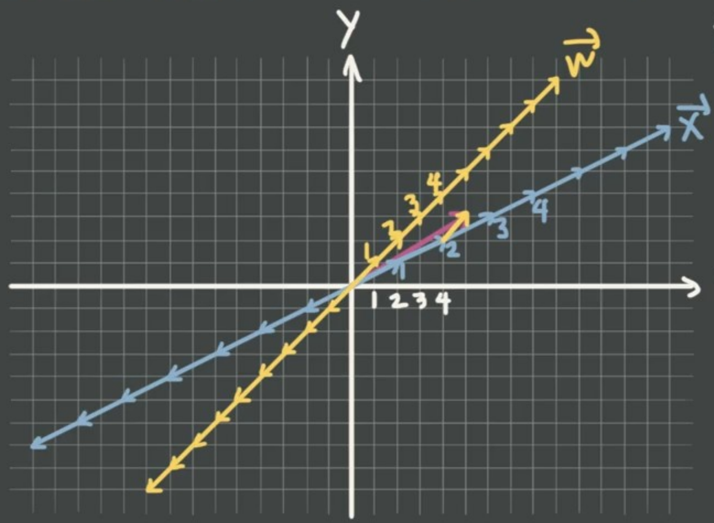
$$\left[\begin{array}{cc|c} 1 & 0 & 13/35 \\ 0 & 1 & 43/35 \end{array}\right] \begin{array}{l} \approx 0.37 \\ \approx 1.23 \end{array}$$

$-7 + 6\left(\frac{43}{35}\right)$

$-\frac{245}{35} + \frac{258}{35} = \frac{13}{35}$

$$\vec{x}^* = \begin{bmatrix} 13/35 \\ 43/35 \end{bmatrix}$$

coordinates in a new basis



$\mathbb{R}^2$

$$i = (1,0)$$
$$j = (0,1)$$

$$B = \{i, j\}$$

$$\vec{v} = (5,3) = 5i + 3j = 2\vec{x} + \vec{w}$$

$$\left.\begin{cases} \vec{x} = (2,1) \\ \vec{w} = (1,1) \end{cases}\right\} B$$

$$[\vec{v}]_B = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\overset{A}{\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}} [\vec{v}]_B = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} 2 & 1 & 5 \\ 1 & 1 & 3 \end{array}\right] \quad \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 2 & 1 & 5 \end{array}\right] \quad \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & -1 & -1 \end{array}\right] \quad \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 1 & 1 \end{array}\right] \quad \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array}\right]$$

$$[\vec{v}]_B = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} \qquad \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4+1 \\ 2+1 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

$$A[\vec{v}]_B = \vec{v}$$
$$A^{-1}A[\vec{v}]_B = A^{-1}\vec{v}$$
$$I[\vec{v}]_B = A^{-1}\vec{v}$$
$$[\vec{v}]_B = A^{-1}\vec{v}$$

$$\left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array}\right] \quad \left[\begin{array}{cc|cc} 1 & 1 & 0 & 1 \\ 2 & 1 & 1 & 0 \end{array}\right] \quad \left[\begin{array}{cc|cc} 1 & 1 & 0 & 1 \\ 0 & -1 & 1 & -2 \end{array}\right] \quad \left[\begin{array}{cc|cc} 1 & 1 & 0 & 1 \\ 0 & 1 & -1 & 2 \end{array}\right] \quad \left[\begin{array}{cc|cc} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 2 \end{array}\right]$$

$$A^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$$

$$[\vec{v}]_B = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Transformation matrix for a basis

$$T(\vec{x}) = A\vec{x}$$

$$T(\vec{x}) = \overset{A}{\begin{bmatrix} 1 & 0 \\ 2 & -2 \end{bmatrix}}\vec{x}$$

$$B = \mathrm{Span}\left(\begin{bmatrix} -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ -3 \end{bmatrix}\right)$$

$$C = \overset{C}{\begin{bmatrix} -1 & 2 \\ 0 & -3 \end{bmatrix}} \qquad C^{-1} = \begin{bmatrix} -1 & -\frac{2}{3} \\ 0 & -\frac{1}{3} \end{bmatrix}$$

$$[T(\vec{x})]_B = M[\vec{x}]_B$$

$$[\vec{x}]_B = (1,3)$$

$$C[\vec{x}]_B = \vec{x}$$

$$M = \begin{bmatrix} -1 & -\frac{2}{3} \\ 0 & -\frac{1}{3} \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 2 & -2 \end{bmatrix}\begin{bmatrix} -1 & 2 \\ 0 & -3 \end{bmatrix}$$

$$M = C^{-1}AC$$

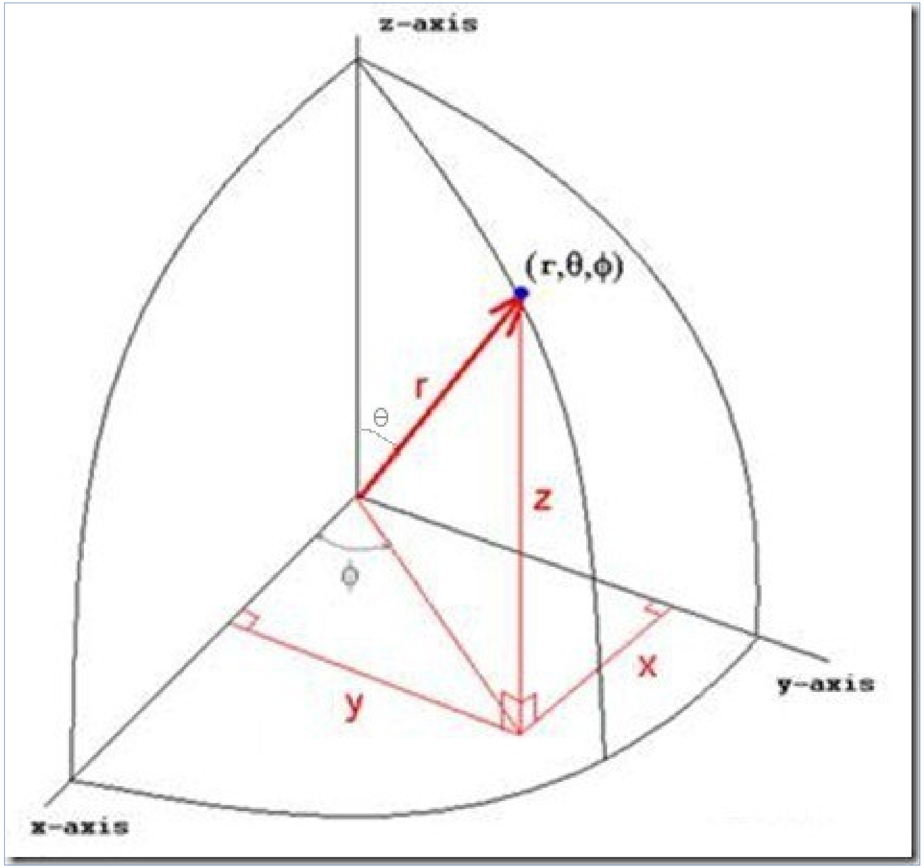
$$= \begin{bmatrix} -1 & -\frac{2}{3} \\ 0 & -\frac{1}{3} \end{bmatrix}\begin{bmatrix} -1 & 2 \\ -2 & 10 \end{bmatrix} = \begin{bmatrix} 1+\frac{4}{3} & -2-\frac{20}{3} \\ 0+\frac{2}{3} & 0-\frac{10}{3} \end{bmatrix} = \begin{bmatrix} \frac{7}{3} & -\frac{26}{3} \\ \frac{2}{3} & -\frac{10}{3} \end{bmatrix}$$

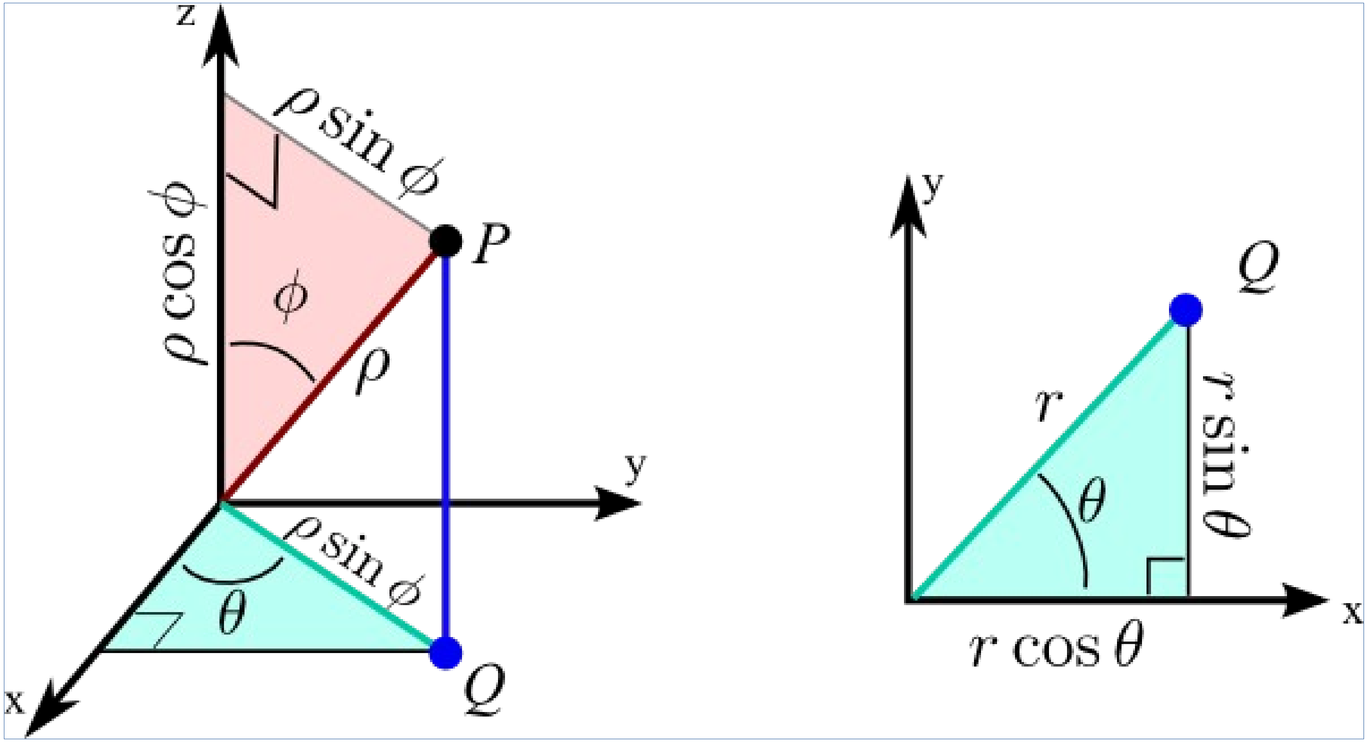
Domain $\qquad\qquad$ codomain

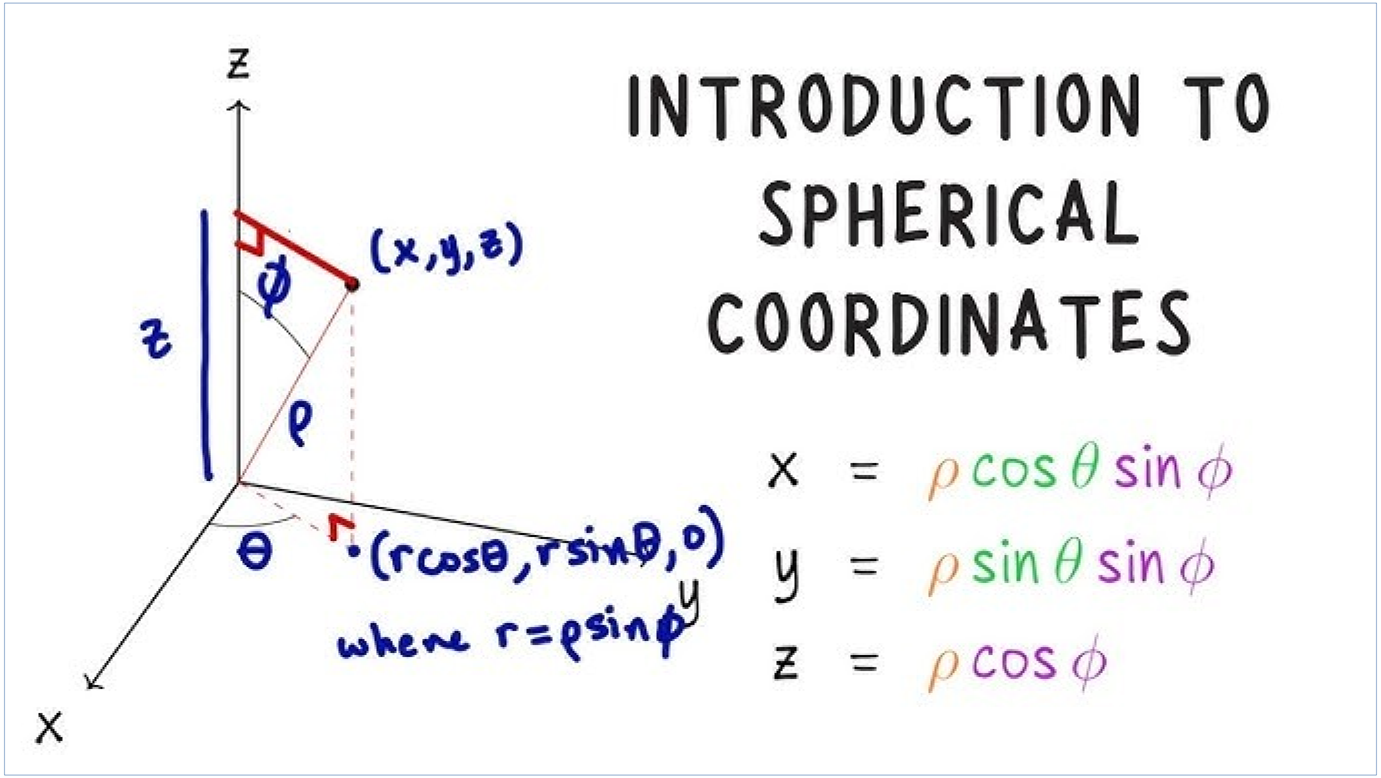
$$\begin{array}{ccc} \vec{x} & \xrightarrow[A]{T} & T(\vec{x}) \\ {\scriptstyle C^{-1}}\downarrow & & \downarrow{\scriptstyle C^{-1}} \\ [\vec{x}]_B & \xrightarrow[M]{T} & [T(\vec{x})]_B \end{array}$$

S.B. $\quad [T(\vec{x})]_B = \frac{1}{3}\begin{bmatrix} 7 & -26 \\ 2 & -10 \end{bmatrix}\begin{bmatrix} 1 \\ 3 \end{bmatrix} \qquad = \frac{1}{3}\begin{bmatrix} 7 & -26 \\ 2 & -10 \end{bmatrix}$

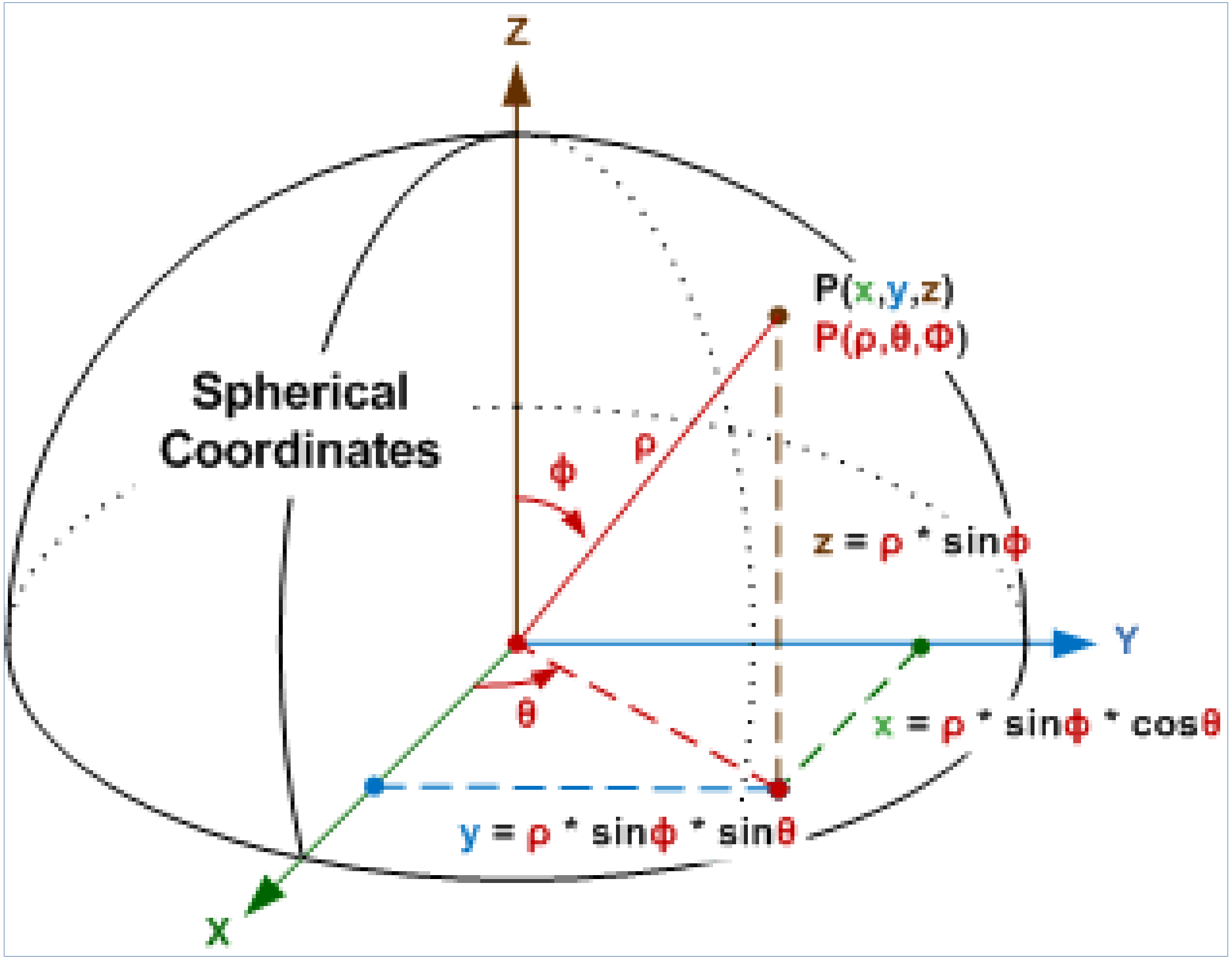
A.B. $\quad = \frac{1}{3}\begin{bmatrix} 7-78 \\ 2-30 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} -71 \\ -28 \end{bmatrix} = \begin{bmatrix} -71/3 \\ -28/3 \end{bmatrix}$


z-axis
(r,θ,ϕ)
r
θ
z
ϕ
x
y
y-axis
x-axis


z
$\rho \sin\phi$
$\rho \cos\phi$
P
ϕ
ρ
y
$\rho \sin\phi$
θ
x
Q
y
Q
r
$r \sin\theta$
θ
x
$r \cos\theta$



$$
\begin{aligned}
x^2 + y^2 + z^2 &= \rho^2 \sin^2 \phi (\cos^2 \theta + \sin^2 \theta) + \rho^2 \cos^2 \phi \\
&= \rho^2 (\sin^2 \phi + \cos^2 \phi) \\
&= \rho^2
\end{aligned}
$$


Z
P(x,y,z)
P(ρ,θ,Φ)
Spherical
Coordinates
ρ
ϕ
z = ρ * sinϕ
Y
θ
x = ρ * sinϕ * cosθ
y = ρ * sinϕ * sinθ
X